

Modeling the trajectory of fireworks

Mathematics HL Internal Assessment

The one element that cannot miss from any New Years Eve or large-scale festival is the pleasure of watching fireworks exploding in various colors and various patterns in the sky. Despite the popularity of fireworks, only few know about the importance of mathematics behind firing an exploding shell. However, mathematics is crucial: an estimate of 17,800 reported fires in the US were caused by fireworks in 2011. Even more, 9,600 people in hospital emergency rooms were treated for fireworks related injuries in the same year (Hall). Any responsible pyrotechnician should, therefore, pay attention to public safety by modeling the precise trajectory of the fired shells. "You need to know your fallout zone," according to Julie Heckman, who is the executive director of the American Pyrotechnic Association (Hepler). The horizontal displacement travelled by a firework rocket and the radius of the explosion are both factors that determine not only the safety of the crowd of spectators, but also whether fallout could cause any damage to the environment. Forests, for example, must not be within the fallout zone as this could even lead to fire. For spectators' safety, display fireworks have a minimum spectator distance of 25m (*UK Firework Review*). The fallout zone, as shown in Figure 1, should be at least 50m in length. During this investigation, it will be assumed that the explosion of the firework has to occur within the fallout zone, in order for it to be safe.

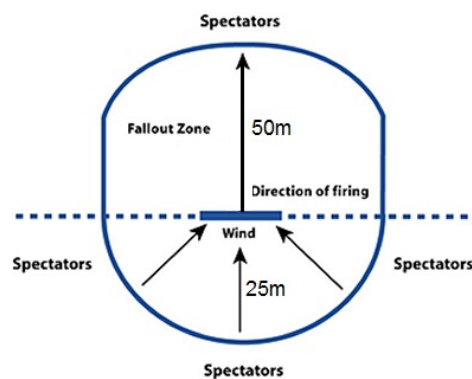


Figure 1.: Safety instructions for setup of display fireworks

Image adopted from: "Display Fireworks Manual 2010." *National Resources Canada*, 16 November 2013: Web. 29.12.2013

The pyrotechnician, of course, also wants to make sure that the fireworks give the maximum visual pleasure, hence the height at which the explosion will occur has to be determined beforehand. Also, a function of height depending on horizontal displacement can be used to plan the choreography for a show, if the functions of height and horizontal displacement are also given in terms of the time t .

When I started this exploration, I was wondering how one would set up a firework show if large display fireworks with different sizes were fired. When I was little, I have always been fascinated by the firework shows at New Year's Eve in my hometown, so I think this exploration is an interesting possibility to explore fireworks from a different perspective. The hypothetical show is supposed to take place at a smaller place for a private occasion, hence it is assumed that there is only enough space for 100m fallout zone. At the end of this exploration, I will be able to determine which angle in combination with which firework I could use for a show under these conditions.

Before attempting to solve the problem laid out in the previous paragraph, the mathematical relations and impacts between the relevant factors have to be studied. The essential purpose of this exploration is to find an equation to model the trajectory of display fireworks of certain sizes and at certain launching angles. The sizes considered are displayed in Table 1. For this, different factors will be considered: the angle of firing is certainly important, and it will be considered in order to be able to determine the relationship between height and time, horizontal displacement and time, and height and horizontal displacement. For part 1, the launching angle will be kept constant, in order to work out the other mathematical relations, and later, the launching angle will also be generalized. A common launching angle used by pyrotechnicians is 75° , not just because it produces good visual results, but also because changing the angle would change the fallout zone (*National Resources Canada*). Different shells with different shell sizes are launched with different initial velocities - initial velocity refers to the speed with which the shell is fired out of the mortar (*thinkquest*). This factor will be varied, and a goal is to find a relationship between the initial velocity and the height and horizontal displacement at which the shell explodes. Furthermore, it is assumed that the shell explodes when it reaches its maximum height. This, of course, is a simplification, but it will allow a more focused approach in answering my main goal. The physical units of velocity, acceleration etc. will not always be mentioned in the mathematical equation. However, units of significant outcomes will be indicated. Wind has a great impact on the trajectory of the fireworks, and any pyrotechnician would have to take it into account. Thus, it will be discussed at certain points during the exploration, however, its influence will not be calculated, as this would exceed the limits of this exploration.

Part 1: Finding the vertical and horizontal displacement depending on the time after launching a certain shell with a certain initial velocity

First, the vertical displacement h vs. time t function - $h(t)$ - will be calculated for different shell sizes. Although the terms distance and displacement have different meanings, they are interchangeable in this exploration. Nevertheless, the term displacement will be used for consistency. Also, the effect of the shell size itself on the trajectory of the firework will not be considered here because acceleration due to gravity is independent of mass. However, shells of different sizes are recommended to be fired

with different initial velocities. Larger shells create greater explosion and they are supposed to explode at greater heights, thus their initial velocity is greater. Initial velocity is defined as v_0 . The initial velocity data for shells with different sizes are shown in Table 1.:

Shell size in mm	Initial velocity v_0 in m/s
20.3	71.6
25.4	80.2
30.5	87.6

Table 1.: Initial velocities of firework shells with different sizes

Source: The Physics of Fireworks."thinkquest.org: Web. 22.12.2013

1.1 Workings with a shell of size 20.3mm



Display firework shells

Source of Image: <http://science.howstuffworks.com/innovation/everyday-innovations/fireworks1.htm>

The shell with size 20.3mm will be chosen as the first example. Also, it is assumed that there is a 100m long fallout zone for this certain show, which means that the horizontal displacement travelled until the explosion must not exceed 100m. Knowing that the launching angle is 75° , the vertical component of the initial velocity can be found easily, as shown in Figure 2. The letter α is used to describes the launching angle:

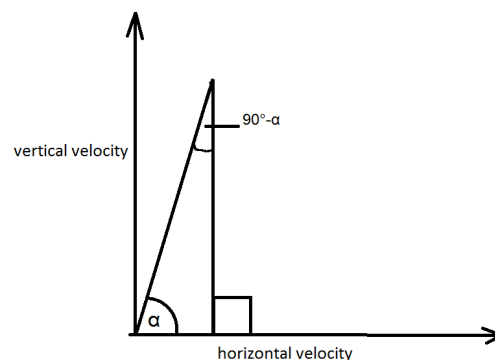


Figure 2.: Component vectors of the initial velocity of a shell fired at 75°

The derivative of the height h of the shell with respect to time t is equal to the vertical velocity:

$$v_{\text{vertical}} = \frac{dh}{dt} = v_0 \cdot \sin(\alpha) = 71.6 \frac{\text{m}}{\text{s}} \cdot \sin(75^\circ) = 69.2 \frac{\text{m}}{\text{s}}, \text{ at } t=0$$

Furthermore, it is known that the vertical acceleration $\frac{d^2h}{dt^2}$ of the shell is -9.81ms^{-2} , since this is the acceleration due to gravity. Integration can be used to find the function $h(t)$:

$$\int -9.81 dt = -9.81t + c = \frac{dh}{dt}$$

$$69.2 = -9.81(0) + c, \text{ hence, } c = 69.2$$

$$\frac{dh}{dt} = -9.81t + 69.2$$

$$\int (-9.81t + 69.2) dt = \frac{-9.81}{2} t^2 + 69.2t + c = h(t)$$

Knowing that $h(0)=0$, we can deduce that $c=0$, hence:

$$h(t) = \frac{-9.81}{2} t^2 + 69.2t$$

where: $h(t)$... in m

t ... in s

This function is graphed in figure 3.

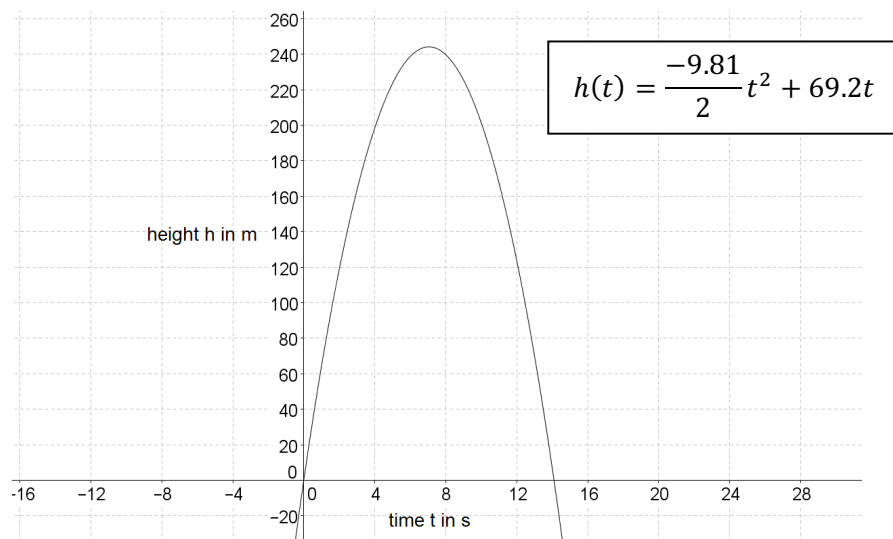


Figure 3.: $h(t)$ when $v_0=71.6\text{m/s}$

The maximum height of the firework can now be determined graphically or algebraically. Since the algebraic way will also be used later, it will be used here to find the coordinates of the maximum of $h(t)$:

$$\frac{dh}{dt} = -9.81t + 69.2$$

Setting the derivative equal to zero and solving for t gives the time of the maximum height value:

$$t = \frac{69.2}{9.81} s = 7.05s, \frac{d^2h}{dt^2} < 0$$

Substituting $t=7.05s$ into $h(t) = \frac{-9.81}{2}t^2 + 69.2t$ will result in:

$$h(7.05) = -(4.905)(49.7) + 69.2(7.05) = 244m$$

Therefore, a shell with 20.3mm size will reach its maximum height of 244m after 7.05s. In order to evaluate whether the firework is within the safety zone at that height, the height h in terms of vertical displacement w - $h(w)$ - needs to be determined. In order to do this, the function of horizontal displacement w in terms of time t - $w(t)$ - will be calculated first. Using figure 2, again:

$$\frac{dw}{dt} = v_0 \cdot \sin(90 - \alpha) = 71.6 \cdot \sin(15^\circ) = 18.5m/s$$

Since there is no constant horizontal force acting on the trajectory of the shell, $w(t)$ can be modeled as a linear function:

$$\int 18.5dt = 18.5t + c$$

since $w(0)=0$, $c=0$. Thus:

$$w(t) = 18.5t$$

The horizontal displacement until the time of explosion is:

$$w(7.05) = 131m$$

In this case, the trajectory would exceed the 100m fallout zone in its moment of explosion. Next, it is useful to find a function that models the height depending on the horizontal displacement $h(w)$, since this could be used to model the trajectory of fireworks. The height for this firework can be expressed in terms of the horizontal displacement by solving $w(t)$ for t and then substituting for t in the $h(t)$ -equation:

$$t = \frac{w}{18.5}$$

$$h(t) = \frac{-9.81}{2}t^2 + 69.2t$$

$$h(w) = \frac{-9.81}{2}\left(\frac{w}{18.5}\right)^2 + 69.2\left(\frac{w}{18.5}\right) = -\frac{109}{7600}w^2 + 3.74w$$

$$h(w) = -\frac{109}{7600}w^2 + 3.74w$$

The resulting function is shown in figure 4.

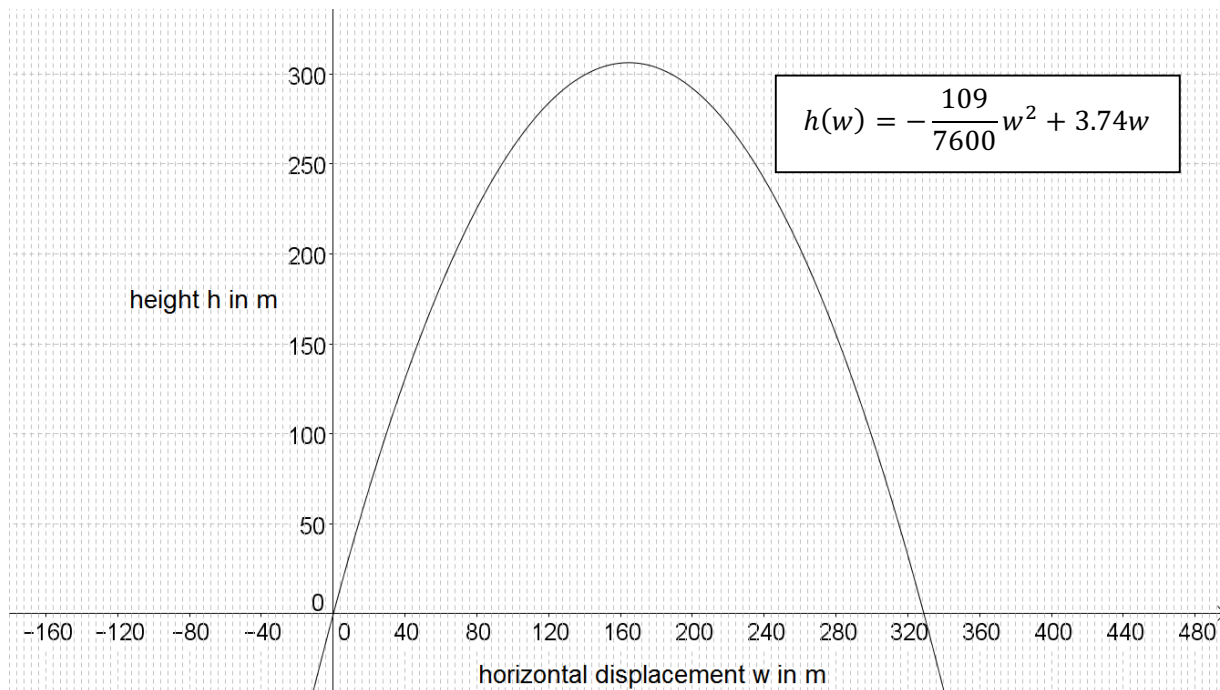


Figure 4.: The height of the shell h in terms of the horizontal displacement w

The graph in figure 4, again, shows that the maximum height is at the point (131,244). As already stated, this height, is not safe because the horizontal displacement at this point exceeds the maximum fallout distance of 100m. It should be noted that after the moment of explosion, the trajectory of the shell will change in an unpredictable direction, thus, it could not be modeled accurately. Figure 4 is therefore only an accurate graph until the explosion occurs.

1.2 Workings with a shell of size 25.4mm and arriving at the general formulas of h(t) and w(t)

A shell of size 25.4mm is launched out of the mortar with an initial velocity v_0 80.2m/s. The vertical velocity at $t=0$ s, therefore, is equal to:

$$\frac{dh}{dt} = 80.2 \cdot \sin(75^\circ) = 77.5 \text{ m/s}$$

By doing the same operations as for the previous shell, $h(t)$ was found, as shown in figure 5:

$$h(t) = \frac{-9.81}{2}t^2 + 77.5t$$

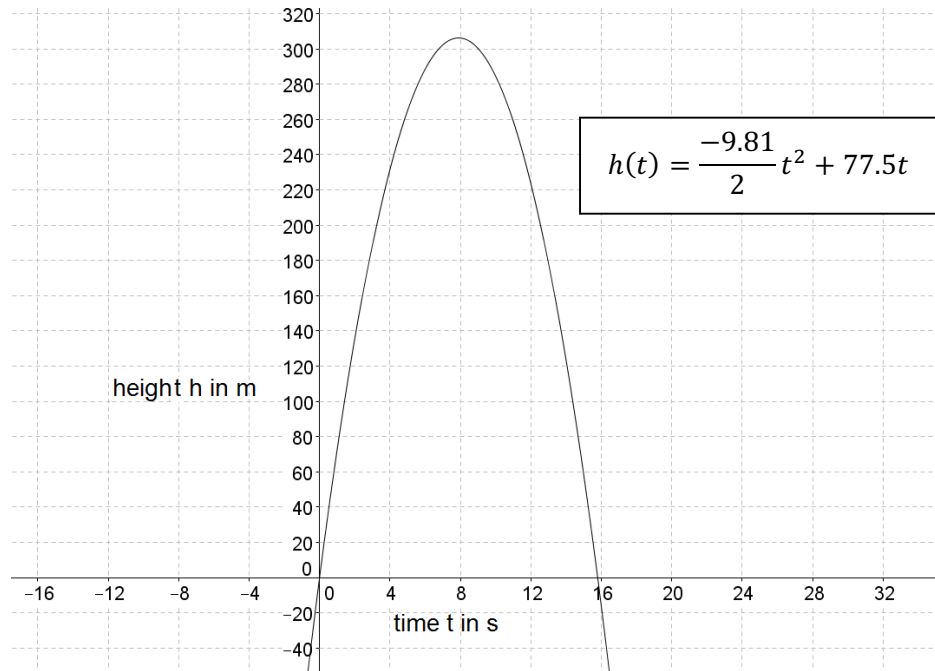


Figure 5.: $h(t)$ for a firework shell of size 25.4mm

The general rule that can be deduced is that for any shell with known initial velocity v_0 and a launching angle of 75° , the height of the shell at any time after launching can be expressed as:

$$h(t) = \frac{-9.81}{2}t^2 + v_0 \sin(75^\circ) t$$

The procedure for finding $w(t)$ was also repeated for this particular shell:

$$w(t) = \sin(15^\circ) \cdot 80.2t = 20.8t$$

In this case, the general formula for $w(t)$ is:

$$w(t) = \sin(15^\circ) t v_0$$

Part 2: modeling the trajectory of shells by finding the relationship between vertical and horizontal displacement

Next, the height h in terms of horizontal displacement w , $h(w)$, for a shell of size 25.4mm will be found, and a general formula for $h(w)$ will be deduced:

$$t = \frac{w}{20.8}$$

Hence:

$$h(w) = \frac{-9.81}{2} \left(\frac{w}{20.8} \right)^2 + 77.5 \left(\frac{w}{20.8} \right) = -\frac{981}{86528} w^2 + 3.726w$$

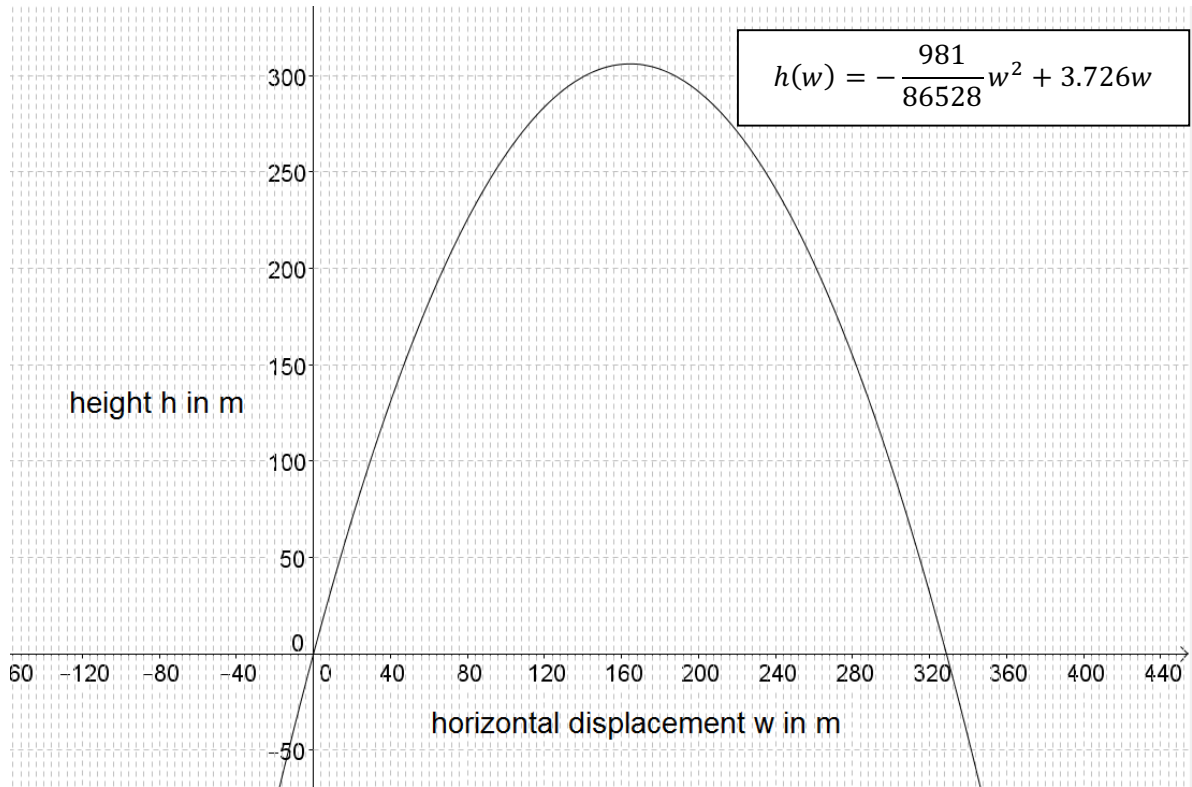


Figure 6.: $h(w)$ for a shell of size 25.4mm

The maximum height will be reached when:

$$\frac{dh}{dw} = -\frac{2 \cdot 981}{86528} w + 3.726 = 0$$

$$w = \frac{(3.726)}{\left(\frac{1962}{86528}\right)} = 164.3m$$

Hence, the maximum height is reached at a horizontal displacement of 164.3m. This would also exceed the 100m safety distance, hence the explosion would have to be timed before the shell reaches its maximum height.

Next, the general formula for $h(w)$ in terms of v_0 and the launching angle α is derived:

$$t = \frac{w}{\sin(90^\circ - \alpha) v_0}$$

$$\begin{aligned}
 h(w) &= \frac{-9.81}{2} \left(\frac{w}{\sin(90^\circ - \alpha) v_0} \right)^2 + \sin(\alpha) v_0 \left(\frac{w}{\sin(90^\circ - \alpha) v_0} \right) \\
 &= \frac{-9.81}{2} \left(\frac{w}{\sin(90^\circ - \alpha) v_0} \right)^2 + \left(\frac{w \sin(\alpha)}{\sin(90^\circ - \alpha)} \right)
 \end{aligned}$$

Thus, for any firework with any launching angle α° :

$$h(w) = \frac{-9.81}{2} \left(\frac{w}{\sin(90^\circ - \alpha) v_0} \right)^2 + \left(\frac{w \sin(\alpha)}{\sin(90^\circ - \alpha)} \right)$$

Using this formula, the trajectory of any shell with any given initial velocity and launching angle can be modeled.

Part 3.: Designing own display firework show

Now, I will come back to my main question, the design of a firework show with a fallout zone of 100m length, where different shells with different sizes should be fired. I want to determine which sizes and at which angles would create the best visual pleasure, so that the firework explodes still within the fallout zone. For this, I will consider which angles one would have to use for the shells, if they explode at their maximum height. Next, I will derive a function that determines the horizontal displacement at which a shell reaches its maximum height. For that, I will first find the first derivative of the height with respect to the horizontal displacement.

$$\frac{dh}{dw} = -9.81 \cdot \frac{w}{\sin^2(90^\circ - \alpha) (v_0)^2} + \left(\frac{\sin(\alpha)}{\sin(90^\circ - \alpha)} \right)$$

Setting $h'(w)$ equal to zero and solving for w results in a function that can be used to calculate w at the maximum height, depending on the initial velocity and the launching angle of the shell:

$$\frac{dh}{dw} = 0 = -9.81 \cdot \frac{w}{\sin^2(90^\circ - \alpha) (v_0)^2} + \left(\frac{\sin(\alpha)}{\sin(90^\circ - \alpha)} \right)$$

$$9.81 \cdot \frac{w}{\sin^2(90^\circ - \alpha) (v_0)^2} = \left(\frac{\sin(\alpha)}{\sin(90^\circ - \alpha)} \right)$$

$$w = \frac{1}{9.81} \cdot \left(\frac{\sin(\alpha)}{\sin(90^\circ - \alpha)} \right) \cdot \sin^2(90^\circ - \alpha) \cdot (v_0)^2$$

$$w = \frac{(v_0)^2}{9.81} \cdot \sin(\alpha) \cdot \sin(90^\circ - \alpha), \quad \frac{d^2h}{dw^2} < 0$$

Using this equation, the value of w can be found and then substituted into the $h(w)$ -function in order to obtain the height of the shell when it explodes. Using this method, a table which lists possible combinations of initial velocity of the shell with different launching angles was created. The horizontal displacement and the heights of the shell at the explosion were calculated for each combination with Microsoft Office Excel. The angles used range from 60° to 85° . An angle of 90° was not considered because shooting the firework straight up would not be safe for the pyrotechnician. The data is shown in Table 2.

Table 2.: The height and horizontal displacement in m at the explosion of firework shells with sizes of 20.3mm, 25.4mm, and 30.5mm at launching angles of 60° , 65° , 70° , 75° , 80° , and 85°

Shell size in mm	Initial velocity of shell in m/s	Launching Angle in $^\circ$	Horizontal displacement until explosion in m	Height at explosion in m
20.3	71.6	60	226.3	196.0
20.3	71.6	65	200.2	214.6
20.3	71.6	70	168.0	230.7
20.3	71.6	75	130.6	243.8
20.3	71.6	80	89.4	253.4
20.3	71.6	85	45.4	259.3
25.4	80.2	60	283.9	245.9
25.4	80.2	65	251.1	269.3
25.4	80.2	70	210.7	289.5
25.4	80.2	75	163.9	305.9
25.4	80.2	80	112.1	317.9
25.4	80.2	85	56.9	325.3
30.5	87.6	60	338.7	293.3
30.5	87.6	65	299.6	321.3
30.5	87.6	70	251.4	345.4
30.5	87.6	75	195.6	364.9
30.5	87.6	80	133.8	379.3
30.5	87.6	85	67.9	388.1

As already discussed in the introduction, the available space for a fallout zone would be 100m. For additional safety, so that the shell does not explode too close to the limits of the fallout zone, another safety zone of 20% is introduced. This means that the maximum horizontal displacement of a shell is limited to 80m. The three fitting constellations under these circumstances are marked in Table 2 with green. These three fireworks are of different sizes and would all be fired at an angle of 85° . For greatest visual pleasure, the firework of size 30.5mm could be fired because it reaches the greatest height and also produces the largest explosion. If fired at the same time, these three different types of shells would create a nice pattern in which they would all explode at different heights and vertical displacements.

Conclusion and Evaluation

This exploration modeled the trajectory of display fireworks with different sizes, and different initial velocities shot at different angles. First, the height depending on the time after launching the firework has been found. With a similar method, the horizontal displacement depending on the time was also determined. These two functions together helped to determine the 2 dimensional trajectory of a certain shell with any initial velocity. It was assumed that the fireworks explode at their maximum height. Even though this is not necessarily the case in reality, this assumption was made for the purpose of this exploration, and because fireworks with digital timers might exceed the costs for a private show. The resulting function $h(w)$ - height in terms of horizontal displacement - can be very useful for planning for the safety of spectators during the show, as it shows whether the shell exceeds the fallout zone at the point of explosion. Furthermore, the $h(w)$ function could also be applied for planning the construction of a larger show with certain visual patterns. The height and horizontal displacement at which one wants to explode a certain firework just has to be substituted into $h(w)$, and then the function can be solved for the initial velocity - for any launching angle. Using that information, one can then also find the time for which the explosion should be set at, using $h(t)$ or $w(t)$. In my example, I assumed the explosion to occur at the maximum height of the firework. To find the horizontal and vertical displacement of the shell at this point, I derived a function for the horizontal displacement when the height is at its maximum. I could then substitute the vertical displacement w into the function $h(w)$ and find h , the height of the firework. In Table 2, I repeated this procedure for a number of angles, ranging from 60° to 85° , and with three different shell sizes of 20.3mm, 25.4mm, and 30.5mm, where shells of each size had certain initial velocities. Due to the limited area available, only three constellations would work for my own show: the three differently sized shells would all be safe at a launching angle of 85° . The fallout zone available would be 100m, however, I introduced an additional safety zone of 20%, which limited the safe horizontal displacement of fireworks to under 80m. Only these three combinations fulfill this requirement. At a launching angle of 85° , a shell with

size 20.3mm would travel 45.4m horizontally and 259.3m vertically. This is certainly the safest combination, because it would be the furthest away from the boundaries of the fallout zone. However, the shell of size 30.5mm would be the most spectacular because it has the largest explosion radius and it would explode at the greatest height of 388.1m. The displacements of the shell with size 25.4mm would fall between the values of the other two. For my show, I would use a combination of these three constellations. This would be enough to have a spectacular firework show.

On the other hand, there are limitations to the mathematics of this exploration. The horizontal as well as vertical displacement is impacted by the force of wind. Although it could be integrated into the formula, wind speed would probably be too unpredictable. Pyrotechnicians have to be informed about current wind directions and wind speeds, and consider these when launching the firework. If the wind is blowing against the direction of launching and it is not too strong, the show would still be safe without changing much because then the fallout would just land further within the fallout zone. However, the show can become dangerous if the wind is blowing in the firework's direction, thereby increasing its horizontal displacement so that it might exceed the fallout zone. In this case, the angle of launching and the timing of the explosion would have to be adjusted. Also, if there is very strong wind, the safest solution would be to cancel the firework display, as the shells trajectory could become greatly unpredictable.

Another major limitation to this exploration is that it does not encounter for air resistance. Certainly, this is another factor that would change the trajectory of the shells. Furthermore, the modeling of the trajectory of the shells is limited to the point where the firework explodes. After that, the exact direction of the fallout becomes very difficult to predict.

Overall, this exploration not just raises the issues of fireworks safety and the responsibilities of anyone working with fireworks, but also how we can use simple mathematics to solve practical problems in daily life. Of course, despite the possible accuracy of these calculations, it should always be consulted with a professional pyrotechnician to ensure complete safety of the show. The results of this investigation are a good theoretical basis for planning a spectacular choreography, however, they should not be used as a substitute for professional experience.

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